

Simulation Report: 4x4 WDM Optical Communication System

1. Project Overview

This report details the design, configuration, and performance analysis of a 4-channel optical communication link. The architecture utilizes Wavelength Division Multiplexing (WDM) to transmit four independent data streams over a single standard optical fiber. The system's performance was evaluated using standard telecommunications metrics, specifically focusing on signal distortion and error rates over distance.

2. System Architecture

The system was engineered into three distinct stages:

- Transmitter Array (Tx):** Consists of four parallel channels. Each channel generates a Pseudo-Random Bit Sequence (PRBS), shaped by an NRZ Pulse Generator, which drives a Mach-Zehnder Modulator. The optical carrier waves are supplied by Continuous Wave (CW) Lasers.
- Optical Link:** The four modulated signals are combined using a 4x1 WDM Multiplexer and transmitted across a 50 km Single-Mode Optical Fiber.
- Receiver Array (Rx):** A 1x4 WDM Demultiplexer separates the signals. Each channel is processed by a PIN Photodiode for optical-to-electrical conversion, smoothed by a Low Pass Bessel Filter, and finalized by a 3R Regenerator to recover the digital sequence.

3. Simulation Parameters

Parameter Category	Specific Values
Carrier Frequencies	193.1 THz, 193.2 THz, 193.3 THz, 193.4 THz
Channel Spacing	100 GHz (0.1 THz)
Transmission Medium	Optical Fiber, Length = 50 km

4. Results and Performance Analysis

The system was analyzed using integrated diagnostic visualizers. Data was captured at the output of the multiplexer and at the final output of Receiver Channel 1.

Optical Spectrum Analysis (OSA)

The OSA connected post-multiplexer confirms the successful aggregation of the four distinct laser frequencies. The spectrum displays four clearly defined signal peaks without severe overlapping, validating the 100 GHz channel spacing and the functionality of the MUX/DEMUX components.

Signal Integrity (BER Analyzer)

The recovered signal was evaluated against the original PRBS data stream to generate an Eye Diagram. The recorded metrics for Channel 1 are:

- **Maximum Q-Factor:** 3.97034
- **Minimum Bit Error Rate (BER):** 2.86963e-05

5. Conclusion and Optimization Strategy

The simulation proves the fundamental architecture of the 4x4 WDM system is functionally sound; signals are successfully multiplexed, transmitted, and demultiplexed. However, the recorded Q-Factor of ~3.97 falls below the standard baseline (typically $Q \geq 6$), and the BER indicates a higher-than-ideal error rate over the 50 km span due to natural fiber attenuation and dispersion.

Proposed Future Work: To optimize this system for deployment or grading, the signal-to-noise ratio must be improved. The next iteration of the simulation should include an Erbium-Doped Fiber Amplifier (EDFA) placed either inline or as a pre-amplifier, or alternatively, an adjustment to the initial CW Laser input power to mitigate the losses accumulated over the 50 km transmission distance.